



Republic of Rwanda
Ministry of Education



QUALITY ASSURANCE APPLICATION

GENQA501

Apply Quality Assurance

Competence

RQF Level: 5

Learning Hours



Credits: 5

Sector: ICT AND MULTIMEDIA

Trade: SOFTWARE DEVELOPMENT

Module Type: General

Curriculum: ICTSWD5001-TVET CERTIFICATE V IN SOFTWARE DEVELOPMENT

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Purpose statement	This general module aims to equip learners with knowledge and skills necessary for applying Quality Assurance. This module is intended to prepare students pursuing TVET Level 5 in software development. Upon completion of this module, learners will be able to analyse system requirements specification, develop test plans, test the system and generate test documentation.						
Learning assumed to be in place	N/A						
Delivery modality	Training delivery			100%	Assessment		Total 100%
	Theoretical content			30%	Formative assessment	30%	50%
	Practical work:			70%		70%	
	Group project and presentation	20%					
	Individual project /Work	50%					
				Summative Assessment		50%	

Elements of Competence and Performance Criteria

Elements of competence	Performance criteria
1. Perform Requirements Analysis	1.1. Terms of reference are properly analysed according to industry quality assurance standards.
	1.2 Requirement specification is effectively examined based on industry standards, project objectives, and stakeholder needs.
	1.3 Inception report is properly analysed based on established criteria and project requirements.
2. Test the System	2.1 Test plan is properly prepared based on specified criteria and industry best practices.
	2.2 Testing environment is properly prepared based on specified criteria and industry best practices.
	2.3 Tests are effectively performed according to the specified parameters and guidelines.
	3.1 Test results are properly consolidated based on test findings.

3. Generate Test Documentation	3.2 Final User Acceptance Testing Report (UAT) is clearly provided based on acceptance criteria.
	3.3 Recommendation Report is properly generated based on comprehensive test cases.

Knowledge	Skills	Attitude
• Description of Quality Assurance(QA) • Differentiation of Quality Assurance and Quality Control • Description of the terms of Reference • Description of stakeholder requirements • Description of test plan • Description of stakeholder expectations • Identifying test tools • Identification of test techniques • Identification of key components of a testing Environment • Classification of quality Assurance	• Analysing Terms of Reference • Examining requirement specification • Generation of findings document • Analysing the inception report • Scheduling test delivery plan • Selecting test techniques • Selecting software testing tools • Setting up and Configuration testing environment • Assessing Acceptance Criteria • Summarizing the key UAT activities conducted • Analysing overall test results • Generating UAT report • Providing Recommendations • Documenting and archiving test artifacts	• Team Player. • Self Confidence. • Detail oriented • Good observer • Creative • Critical thinking • Problem solver • Self-motivated • Continuous learner • Innovative • Teamwork • Hardworking • Speak with Relevance

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Perform Requirements Analysis 2. Test the System 3. Generate Test Documentation
Learning outcome 1: Perform Requirements Analysis	Learning hours: 10
Indicative content	
• Introduction to Quality Assurance ✓ Describing Quality Assurance(QA)  Definition  Stages/Processes  Main Types  Standards  Methods  Benefits ✓ Classification of quality assurance  Industry-Specific QA  Process and Product QA  Internal vs. External QA  Manual vs. Automated QA ✓ Quality Assurance Vs Quality Control ✓ Identify tools and techniques • Analysing Terms of Reference (TOR) ✓ Describing Terms of Reference  Definition  Purpose  Elements of TOR  Importance  Test cases ✓ Document a TOR analysis report • Examining requirement specification ✓ Review the entire document	

✓ Understand stakeholder requirements  Functional  Non-Functional ✓ Verify consistency and completeness ✓ Identify dependencies and interactions ✓ Generate findings document • Analysing inception report ✓ Perform document review ✓ Understand stakeholder expectations ✓ Assess clarity and completeness ✓ Evaluate feasibility and risks ✓ Document assumptions and constraints ✓ Document analysis findings	
Resources required for the learning outcome	
Equipment	• Computer • Projector • Printer
Materials	• Electricity
Tools	• Check list
Facilitation techniques	• Brainstorming • Group discussion • Trainer guided • Jig saw
Formative assessment methods /(CAT)	• Oral assessment • Written assessment • Practical assessment

Indicative content

- **Preparation of Test Plan**
 - ✓ Description of test plan
 - ✚ Definition
 - ✚ Types
 - ✚ Components
 - ✓ Identify test tools
 - ✓ Identify test techniques
 - ✓ Identify test cases
 - ✓ Identify test criteria
 - ✓ Allocate resources
 - ✓ Prepare Test Data
 - ✓ Prepare Test Execution Scripts
 - ✓ Schedule test delivery plan
- **Preparation of Testing Environment**
 - ✓ Identifying key components of a Testing Environment
 - ✚ Hardware Infrastructure
 - ✚ Software Infrastructure
 - ✚ Dependencies needed for testing
 - ✓ Selecting test techniques
 - ✚ Manual testing
 - ✚ Automated Testing
 - ✚ Exploratory Testing
 - ✚ Parallel Testing
 - ✚ Matching Test Techniques with system functionalities
 - ✓ Select testing tools
 - ✚ Software
 - ✚ Hardware
 - ✓ Set up and configure testing environment
- **Perform Testing**
 - ✓ Review test plan
 - ✓ Select test cases
 - ✓ Process test
 - ✓ Record defects
 - ✓ Monitor test progress

✓ Document test results	
Resources required for the learning outcome	
Equipment	• Computer • Smartphone • Tablet
Materials	• Pen • Notebook
Tools	• Microsoft Visio, • LucidChart, • DrawIO
Facilitation techniques	• Brainstorming • Group discussion • Demonstration • Practical exercise
Formative assessment methods /(CAT)	• Oral assessment • Written assessment • Practical assessment

Learning outcome 3: Generate Test Documentation

Learning hours: 13

Indicative content

- **Consolidation of test results**
 - ✓ Collecting test data
 -  Collecting test artifacts
 -  Compiling test execution data
 - ✓ Analysing and evaluating test results
 -  Reviewing Test Logs and Reports
 -  Summarizing Test Findings
 -  Prioritizing Defects
 -  Generating Test Metrics
 - ✓ Create test summary report
- **Providing User Acceptance Testing Report**
 - ✓ Assess Acceptance Criteria
 - ✓ Summarizing the key UAT activities conducted
 -  Executing test case and results(passed/failed/blocked)
 -  Identifying usability testing outcomes (user feedback, pain points)
 -  Measuring performance testing (speed, stability)
 -  Security vulnerability findings
 - ✓ Analysing overall test results
 - ✓ Generate UAT report
- **Generating recommendation Report**
 - ✓ Provide Recommendations
 -  Proposing changes.
 -  Present supporting evidence
 -  Outline action steps
 - ✓ Include Benefits and Impact
 - ✓ Document and archiving test artifacts

Resources required for the learning outcome

Equipment	• Computer
Materials	• Pens • Notebook
Tools	• Version control systems: Git, Subversion • Screenshot/recording tools: Snagit, Screencastify

Facilitation techniques	• Brainstorming • Group discussion • Demonstration • Practical exercise
Formative assessment methods /(CAT)	• Oral assessment • Written assessment • Practical assessment

References:

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- MODULES FOR TOTAL QUALITY MANAGEMENT (TQM) – Hermon Quality Solutions*. (n.d.). *Www.hermonqualitysolutions.com*. Retrieved April 27, 2024, from
<https://www.hermonqualitysolutions.com/vii-modules-for-total-quality-management-tqm/>
- Sajith. (2023, July 11). *Understanding the Four Types of Quality Assurance for Exceptional Results*. Medium. <https://medium.com/@sajiveva1112000/understanding-the-four-types-of-quality-assurance-for-exceptional-results-bef3168060fc>